Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2010-0233	Issued	2010-06-07

Profile Radiography Report - 4-RA-104 and 4-LA-104

Scope and Responsibilities

This report details the profile radiography performed on insulated lines 4-RA-104 and 4-LA-104 within the ATU-100 at the Hamilton Plant. The inspection was triggered by QSCP-INS-2010-021, which identified damaged jacketing and wet insulation during a 5-yearly external visual inspection, leading to a recommendation for CUI investigation. Marc Tremblay, acting as the RT technician (SNT-TC-1A RT Level II) for Halward Inspection Services Ltd., was responsible for the radiographic exposures and initial film interpretation.

The examination was conducted to assess the external condition of the pipe walls under insulation without stripping. All radiographic testing was performed in accordance with ASME BPVC Section V using the HIS-IR-07 exposure device. Thickness evaluations and scanning requirements followed the protocols established in QSCP-NDT-001 Rev 1, ensuring that CMLs were positioned at fittings and elbows to capture potential localized degradation.

Inspection Findings - 4-RA-104

Line 4-RA-104 is the rich side L/R exchanger piping, constructed of A106 Gr.B with an OD of 114.3 mm and a nominal thickness of 8.56 mm. The design limits for this circuit are 862 kPa(g) and 127 °C. Profile radiography indicated wall loss at two distinct locations beneath the insulation. These areas of thinning were evaluated against the retirement thickness (t-min) of 3.81 mm.

The detected wall loss requires further assessment to determine the corrosion rate and remaining life. Per site rules, the remaining life is calculated as (t - t-min) divided by the larger of the long-term or short-term corrosion rates. Because this line is part of the amine circuit, it is subject to a Class 2 maximum thickness interval of 10 years, though the current findings may necessitate a reduced inspection frequency.

Inspection Findings - 4-LA-104

Line 4-LA-104 serves the lean side of the L/R exchanger, consisting of A106 Gr.B material with an OD of 114.3 mm and a nominal thickness of 6.02 mm. The design parameters are 2758 kPa(g) and 138 °C. Radiographic profiles of this line showed only superficial external pitting, with depths measured at <= 0.5 mm. This pitting does not currently approach the retirement thickness (t-min) of 3.18 mm.

The superficial nature of the pitting on 4-LA-104 indicates that the pressure boundary remains well above the mandatory retirement threshold. As a Class 2 line, the next thickness reading is due at no more than half the remaining life or within the 10-year maximum interval. The external visual inspection interval remains at 5 years as per the site requirements for amine service.

Summary of Results

The profile radiography confirmed that the wet insulation noted in QSCP-INS-2010-021 has resulted in active CUI on 4-RA-104, while 4-LA-104 remains largely unaffected by significant metal loss. All measurements were obtained using gauges with waveform display as mandated by QSCP-NDT-001 Rev 1 to ensure accuracy in the presence of external pitting.

Marc Tremblay has completed the film interpretation for all shots taken with HIS-IR-07. The data indicates that while 4-LA-104 is fit for continued service, the two locations of wall loss on 4-RA-104 must be monitored. Any reading found below the specified t-min during subsequent ultrasonic verification must be raised as an NCR in accordance with the Hamilton Plant asset integrity rules.

Sign-off	Name	Function	Date	Signature
Prepared	Marc Tremblay	RT technician, SNT-TC-1A RT Level II	2010-06-07	_____